RESEARCH LETTER

Channel attention in few-label hyperspectral CNNs: an equivalence-tested comparison

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ABSTRACT
Channel-attention blocks are often added to lightweight networks for hyperspectral image classification, with gains reported on random pixel splits. We test whether such gains are real and whether they transfer. Seven channel-modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of SNSM, a dispersion-aware block used as a controlled probe) are compared on one backbone with one training protocol, no test-based model selection, up to fifteen paired seeds, Holm-corrected Wilcoxon tests and equivalence tests with a margin of ± 1 point of average accuracy (AA). On random 5% splits, blocks differ only on Indian Pines, where the dispersion-based gates exceed the others by 1.8–3.8 AA points through two classes with a single training pixel. Once every class has at least 15 training pixels, in random or standard splits, all blocks are equivalent within ± 1 point. Imposing the same scarcity on two other scenes does not reproduce the gap, and on patch-disjoint splits accuracy falls by 32–34 points while most differences remain non-significant. Class counts and evaluation protocol, not the choice of attention block, determine reported accuracy.

KEYWORDS
Hyperspectral image classification; channel attention; equivalence testing; spatial leakage; few labels

1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC) layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hy-

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perspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). This leakage has been surveyed (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare whole architectures. They do not test whether the *choice of channel-attention block* matters, and a block that "brings no significant gain" is usually reported without showing that its effect is actually small.

Our contributions are:

- a controlled comparison of seven channel-modulation blocks on one backbone, with identical inputs and schedule, no test-based model selection and paired seeds, in which the absence of a gain is tested directly with equivalence tests, under six protocols that vary class counts and spatial overlap (Sections 2–3);
- the finding that the blocks are equivalent within ± 1 AA point as soon as every class has 15 training pixels; the one large difference, on random Indian Pines splits, comes from two single-pixel classes, does not reappear under the same scarcity on other scenes, and is not consistently present when patches are disjoint (Section 4);
- a recommendation for evaluating new blocks: report the training pixels of the smallest classes, use equivalence tests rather than non-significance, and repeat the comparison on patch-disjoint splits.

2. Blocks compared

All blocks are inserted in the same backbone. A 3×3 convolution maps the B input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing $\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x})))$ and $\mathbf{y} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x})$, where $\mathcal{M}$ is the block under study ($\mathcal{M} =$ identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, 7×7 spatial gate) output a sigmoid scale per channel; GCT uses its ℓ_2 embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

SNSM as a probe. To test which channel statistic matters, we also include spectral-neighbourhood self-modulation (SNSM), a block whose inputs can be switched on and off. For every channel c it computes the spatial mean, the spatial variance and the difference of the neighbouring channels' means, standardises each across channels, and filters the three with two one-dimensional convolutions along the channel axis (8 hidden channels, kernel 3, dilation 2) into a scale $\gamma_c = 1 + \frac{1}{2}\tanh(z_{1,c})$ and a shift $\beta_c = z_{2,c}$, so that $\mathcal{M}(\mathbf{h})_c = \gamma_c \mathbf{h}_c + \beta_c$. The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block. SNSM (γ) drops the shift. The block diagram and the code are in the supplement and repository.

Table 1. Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

Scene	Sensor	Size (pixels)	Bands	GSD (m)	Classes	Labelled	Train/test (random)
Indian Pines (IP)	AVIRIS	145×145	200	20	16	10,249	505/9,744
Pavia University (PU)	ROSIS	610×340	103	1.3	9	42,776	2,135/40,641
Kennedy Space Center (KSC)	AVIRIS	512×614	176	18	13	5,211	256/4,955
Botswana (BW)	Hyperion	1476×256	145	30	14	3,248	156/3,092

3. Data and protocols

Data. We use Indian Pines (IP), Pavia University (PU), Kennedy Space Center (KSC) and Botswana (BW) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020) (Table 1). Each band is smoothed with a 5×5 Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch. *Note on KSC:* the distributed file stores small negative values (-27 to -1) as unsigned 16-bit integers, which read as 65,509–65,535 and dominate the PCA. Kakde et al. (2018) treated them as saturation and set them to zero; the DeepHyperX loader of Audebert, Le Saux, and Lefevre (2019) keeps them. We read the file as signed integers; read as unsigned, seven of our CNN variants reach only 77–92% AA on KSC instead of 99.2–99.5%.

Protocols. *Random 5%*: for each class, 5% of the labelled pixels (at least one) train and the rest test, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC) and 156/3,092 (BW) pixels; Alfalfa, Grass-pasture-mowed and Oats then have two, one and one training pixels. *Smallest 15* (IP): the same split with only those three classes raised to 15 training pixels. *Random 50/15* (IP): a random split with the class counts of the standard split, 15 training pixels for those three classes and 50 for each other class (695/9,554). *Standard* (IP): the fixed training and test sets distributed with SpectralFormer (Hong et al. 2022) (695/9,671), whose training pixels come from separate regions but whose test patches still overlap training patches at 15×15. *Smallest 1/1/2* (PU, KSC): the random 5% split with the three smallest classes of each scene reduced to 1, 1 and 2 training pixels, as in IP. *Patch-disjoint* (PU, KSC): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 1); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain. This rule leaves no test pixels for five IP classes, and the standard PU and Houston splits were no longer available, so these two scenes carry the patch-disjoint evaluation.

Training and statistics. Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate 10^{-3}, weight decay 10^{-4}), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once. All eight backbone variants use 15 seeds (42–56), except on the random 5% splits of PU, KSC and BW (five seeds); on the standard split the seed changes only the initialisation and batch order. For each of the 8 settings we compare SNSM (γ+β) with the seven other blocks by paired same-seed differences of AA. Differences are tested with two-sided Wilcoxon signed-rank tests, Holm-corrected within each setting; across all 56 comparisons, Benjamini–Hochberg control of the false discovery rate at 5% gives 16 discoveries, against 18 with the per-setting Holm correction. Equivalence is tested with paired two one-sided t-tests (TOST, $\alpha = 0.05$): a difference is equivalent



Figure 1. Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose 15×15 patch would overlap a training patch are excluded.

Table 2. Random 5% splits: OA and AA (%, mean ± std) of the eight backbone variants (15 seeds on IP, five elsewhere). Best mean per column in bold. External baselines (HybridSN, SSFTT) and an FC (γ+β) gate are in the supplement.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	96.28±0.79	89.49±2.44	99.65±0.34	99.51±0.36	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE (γ)	**96.34±0.62**	90.04±2.18	99.61±0.28	99.43±0.38	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA (γ)	96.28±0.68	90.87±2.53	99.61±0.47	99.40±0.66	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.19±0.60	91.58±1.49	99.54±0.56	99.23±0.83	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
GCT	96.28±0.80	90.07±2.34	**99.69±0.26**	**99.55±0.28**	99.58±0.08	98.99±0.18	99.11±0.56	98.88±0.91
SRM	96.20±0.71	93.04±2.14	99.69±0.36	99.51±0.49	99.55±0.07	98.92±0.17	98.87±0.63	98.67±0.97
SNSM (γ)	96.17±0.73	92.89±2.00	99.69±0.26	99.50±0.39	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.27±0.67	**93.33±1.68**	99.63±0.55	99.45±0.73	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03

if its 90% t interval lies within ±1 AA point, less than half the median seed-to-seed standard deviation of AA on random IP splits (2.2 points). Runs used PyTorch (Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds, which ran on an RTX 3050 laptop GPU; each paired comparison uses one GPU type.

4. Results

Random 5% splits. On PU, KSC and BW the eight variants lie within 0.32 AA points of each other (Table 2). Only IP separates them: SNSM (γ+β) is significantly better than the plain CNN, SE, ECA, CBAM and GCT, by 1.75–3.84 points, but not than SRM or SNSM (γ) (Table 4), while the overall accuracy of all eight differs by only 0.17 points. The gap lies almost entirely in the two classes with a single training pixel and 19 and 27 test pixels: SRM and the SNSM variants reach 80.0–84.4% on

Table 3. Indian Pines: AA (%, mean ± std over 15 seeds) of the eight backbone variants under four protocols. The last row is the range of the block means.

Block	Random 5%	Random, smallest 15	Random, 50/15	Standard split
Plain CNN	89.49±2.44	96.03±0.76	97.83±0.52	97.24±0.23
SE (γ)	90.04±2.18	95.93±0.64	**97.90±0.44**	97.25±0.26
ECA (γ)	90.87±2.53	96.10±0.76	97.79±0.49	97.15±0.22
CBAM	91.58±1.49	95.83±0.67	97.78±0.51	97.45±0.29
GCT	90.07±2.34	96.07±0.75	97.82±0.53	97.23±0.22
SRM	93.04±2.14	96.01±0.81	97.76±0.57	**97.50±0.22**
SNSM (γ)	92.89±2.00	96.13±0.72	97.77±0.54	97.37±0.22
SNSM (γ+β)	**93.33±1.68**	**96.43±0.66**	97.82±0.43	97.48±0.26
Range of block means	3.84	0.60	0.14	0.35

Table 4. Paired AA differences SNSM (γ+β) − block (points). *: Holm-corrected Wilcoxon $p < 0.05$ within the row; †: equivalent within ±1 point (paired TOST, $\alpha = 0.05$). 95% intervals and OA differences are in the supplement.

Setting	Plain	SE	ECA	CBAM	GCT	SRM	SNSM (γ)	Seeds
IP, random 5%	+3.84*	+3.29*	+2.45*	+1.75*	+3.26*	+0.29	+0.44	15
IP, random 50/15	-0.01†	-0.08†	+0.04†	+0.05†	+0.01†	+0.07†	+0.06†	15
IP, smallest 15	+0.39*†	+0.50*†	+0.33*†	+0.60*†	+0.35*†	+0.41*†	+0.30*†	15
IP, standard	+0.24†	+0.23†	+0.33*†	+0.04†	+0.25†	-0.02†	+0.12†	15
PU, smallest 1/1/2	-0.59	-0.74	-1.19*	-0.30†	-1.14	-0.55	-1.59*	15
KSC, smallest 1/1/2	-0.24†	+0.79	+0.39	+0.54	-0.21†	+1.41	+0.03	15
PU, patch-disjoint	+0.31	-0.47	-0.39	+1.03	+0.92	+1.33	+0.03	15
KSC, patch-disjoint	+3.46*	-0.19	+1.86	+0.91	+3.51*	+4.25*	+1.40	15

Grass-pasture-mowed and 66.3–77.2% on Oats, against 62.7–78.5% and 36.8–56.1% for the other blocks, whereas on Alfalfa (two training pixels) all blocks lie within 3.3 points and their mean over the other 13 classes differs by less than one point.

Class counts. Raising only the three smallest IP classes to 15 training pixels shrinks the spread of block means from 3.84 to 0.60 points (Table 3). SNSM (γ+β) keeps a small, significant edge of 0.30–0.60 points over every block, but every one of these differences is also equivalent within ±1 point (Table 4). With 15–50 training pixels per class, in the random 50/15 split or the standard split, the block means span 0.14 and 0.35 points and all fourteen comparisons are equivalent within ±1 point; only the difference against ECA on the standard split (0.33 points) is also significant. Because the random 50/15 split keeps the random geometry and changes only the class counts, the IP gap is an effect of the single-pixel classes, not of spatial leakage.

Scarcity elsewhere. Reducing the three smallest classes of PU and KSC to 1, 1 and 2 training pixels lowers AA by about 15 and 7 points, but the dispersion-based gates do not gain (Table 4): on PU, SNSM (γ+β) has the lowest mean AA and is significantly worse than ECA and SNSM (γ); on KSC no difference is significant. Per-block AA is in the supplement.

Patch-disjoint splits. Removing patch overlap lowers AA by 32.0 points on PU and 34.3 on KSC, averaged over blocks, and OA to 55–57% and 74–77% (Table 5). Block means span 1.80 (PU) and 4.44 (KSC) points, but the seed-to-seed standard deviation is 4.4–7.4 points and the median 95% interval of a paired difference is 4.0 (PU) and 3.9 (KSC) points wide, so differences of about two points can be neither confirmed nor excluded, and no comparison is equivalent within ±1 point. On PU no difference is significant; on KSC, SNSM (γ+β) is significantly better than the plain CNN, GCT and SRM, but not than SE, ECA, CBAM or SNSM (γ), and SE has the highest mean AA (Figure 2). The ranking also changes between protocols: SRM, tied

Table 5. PU and KSC: AA under random 5% splits (five seeds) and OA and AA under patch-disjoint splits (15 seeds) (%, mean ± std). The last row is the range of the block means.

Block	PU			KSC		
	AA, random	OA, disjoint	AA, disjoint	AA, random	OA, disjoint	AA, disjoint
Plain CNN	99.01±0.16	55.31±9.82	67.03±5.03	99.51±0.36	75.13±4.97	63.55±6.41
SE (γ)	98.80±0.56	55.91±9.73	**67.81±4.39**	99.43±0.38	76.79±4.44	**67.20±6.66**
ECA (γ)	99.00±0.20	56.30±9.17	67.73±4.43	99.40±0.66	76.40±4.81	65.15±5.86
CBAM	**99.11±0.18**	56.11±10.31	66.31±4.80	99.23±0.83	76.68±4.74	66.10±7.39
GCT	98.99±0.18	54.59±9.68	66.42±4.57	**99.55±0.28**	75.20±5.16	63.49±6.80
SRM	98.92±0.17	56.02±9.99	66.00±4.73	99.51±0.49	74.36±4.22	62.75±6.38
SNSM (γ)	98.91±0.16	55.41±9.44	67.31±4.57	99.50±0.39	76.86±4.19	65.60±7.15
SNSM (γ+β)	98.93±0.16	**56.80±11.83**	67.34±4.95	99.45±0.73	**77.11±3.54**	67.01±6.00
Range of block means	0.31	2.21	1.80	0.32	2.75	4.44

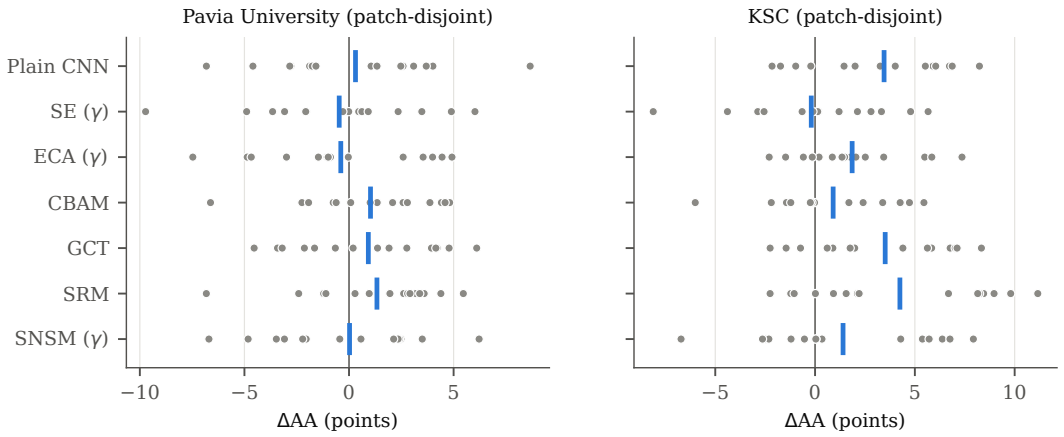


Figure 2. Paired same-seed AA differences SNSM (γ+β) − block on the patch-disjoint splits. Grey dots are the 15 seeds; the blue bar is their mean.

best on random 5% IP, is the lowest on patch-disjoint KSC.

Which statistic matters. Removing inputs from the SNSM probe on random 5% IP (five seeds; Table 6) shows that the variance carries the gap: with the mean alone AA falls by 1.16 points, and adding the gradient to the mean does not recover it; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM and SRM, the only other block that gates on dispersion. All blocks add at most a few thousand parameters; SNSM's many small operations make its batch-1 latency about 3.7 times that of the plain CNN (supplement).

5. Discussion and limitations

Across four scenes and six protocols, the choice of channel-attention block produced a large and consistent difference in one setting only: random 5% splits of Indian Pines, where two classes have a single training pixel. There, gates that see channel dispersion recognised those classes more often. With 15 or more training pixels per class, all blocks were equivalent within ±1 AA point, and the same scarcity imposed on Pavia University and KSC did not favour dispersion-based gates. Spatial leakage changes the level of accuracy more than the ranking: without patch overlap every block lost more than 30 AA points, and the remaining differences were mostly not significant. A gain reported for a new block on a random split should therefore be checked for its dependence on the smallest classes, tested for equivalence rather than for significance

Table 6. SNSM (γ+β) probe ablation on IP, random 5% splits (AA, %; five seeds). ΔAA is the paired difference from the full block; wins count seeds on which the variant is better.

Descriptors	AA	ΔAA (wins/5)
mean, var, grad (SNSM)	94.16±1.07	–
mean	93.01±0.72	-1.16 (1/5)
mean, var	94.55±1.67	+0.39 (4/5)
mean, grad	93.37±1.09	-0.80 (1/5)
mean, var, grad; default init.	92.67±1.15	-1.49 (0/5)

alone, and repeated on patch-disjoint splits.

Several limitations apply. All blocks share one shallow backbone, one patch size, one feature pipeline and a fixed schedule; none was tuned, and deeper networks may use attention differently. PCA is fitted on all pixels of each scene without labels. The four scenes are long-standing benchmarks. On the patch-disjoint splits, 15 seeds can only exclude differences larger than about two points, and the construction adds a spatial shift between training and test data; the standard IP split still lets patches overlap. The scarcity test reduces the smallest classes of PU and KSC, which may be easier than the scarce IP classes.

6. Conclusion

In a lightweight few-label hyperspectral CNN, seven channel-attention blocks were equivalent within ±1 point of average accuracy whenever every class had at least 15 training pixels. The one large difference, 2–4 points in favour of dispersion-based gates on random Indian Pines splits, came from two single-pixel classes, did not appear when the same scarcity was imposed on other scenes, and was mostly not significant on patch-disjoint splits, where accuracy dropped by 32–34 points. Class counts and evaluation protocol, not the attention block, determine reported accuracy.

Disclosure statement

The authors report there are no competing interests to declare.

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Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results, the supplement and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

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